$$\overrightarrow{OC} = z_1 e^{i\frac{4\pi}{3}} = z_2 e^{i\frac{2\pi}{3}} = -2e^{i\frac{2\pi}{3}}$$

$$= -2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$$

$$= -2\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 1 - i\sqrt{3}\,.$$

Therefore, $z_2 = -2$, and $z_3 = 1 - i\sqrt{3}$.

EXERCISE 2.8

1. If $\omega \neq 1$ is a cube root of unity, show that $\dfrac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2} + \dfrac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2} = -1$.

2. Show that $\left(\dfrac{\sqrt{3}}{2} + \dfrac{i}{2}\right)^5 + \left(\dfrac{\sqrt{3}}{2} - \dfrac{i}{2}\right)^5 = -\sqrt{3}$.

3. Find the value of $\left(\dfrac{1+\sin\frac{\pi}{10} + i\cos\frac{\pi}{10}}{1+\sin\frac{\pi}{10} - i\cos\frac{\pi}{10}}\right)^{10}$.

4. If $2\cos\alpha = x + \dfrac{1}{x}$ and $2\cos\beta = y + \dfrac{1}{y}$, show that

   (i) $\dfrac{x}{y} + \dfrac{y}{x} = 2\cos(\alpha - \beta)$ (ii) $xy - \dfrac{1}{xy} = 2i\sin(\alpha + \beta)$

   (iii) $\dfrac{x^m}{y^n} - \dfrac{y^n}{x^m} = 2i\sin(m\alpha - n\beta)$ (iv) $x^m y^n + \dfrac{1}{x^m y^n} = 2\cos(m\alpha + n\beta)$.

5. Solve the equation $z^3 + 27 = 0$.

6. If $\omega \neq 1$ is a cube root of unity, show that the roots of the equation $(z-1)^3 + 8 = 0$ are $-1,\ 1-2\omega,\ 1-2\omega^2$.

7. Find the value of $\sum_{k=1}^{8}\left(\cos\dfrac{2k\pi}{9} + i\sin\dfrac{2k\pi}{9}\right)$.

8. If $\omega \neq 1$ is a cube root of unity, show that

   (i) $(1-\omega+\omega^2)^6 + (1+\omega-\omega^2)^6 = 128$.

   (ii) $(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8)\cdots(1+\omega^{2^{11}}) = 1$.

9. If $z = 2 - 2i$, find the rotation of z by θ radians in the counter clockwise direction about the origin when

   (i) $\theta = \dfrac{\pi}{3}$ (ii) $\theta = \dfrac{2\pi}{3}$ (iii) $\theta = \dfrac{3\pi}{2}$.

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